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Implementation of a novel compliant transmission in a 3 DoF robotic testbench for collaborative hand guiding tasks

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Abstract

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Nomenclature

Symbols

$\boldsymbol{q}$	Joint position vector	[rad]
K_s	Spring stiffness	[Nm/rad]

Abbreviations

CAN	Controller Area Network
SET	Series Elastic Transmission

Chapter 1

Introduction

1.1 Background and Motivation

1.1.1 Musculoskeletal disorders and ergonomic work environments

Non-ergonomic work environments place strain on the human body, which can cause painful complications and limit the worker's ability to perform tasks optimally. Ergonomics can be defined as the study of work, where the goal is to fit tasks and jobs to workers' capabilities, rather than forcing workers to adapt to the job. An ergonomic work environment therefore implements task-specific strategies to prevent reduced work ability. How these complications arise, which parts of the body are affected, and how they can be prevented are explained in this section.

The musculoskeletal system contains muscles, bones, joints and connective tissues like tendons and ligaments. Musculoskeletal disorders (MSDs) can therefore be defined as injuries of the muscles, nerves, tendons, joints, cartilage, and spinal discs. When these injuries are sustained or worsened by exposure to risk factors in the workplace, they are also known as work-related musculoskeletal disorders (WMSDs). These disorders can go from minor aches to severe pain and chronic illness requiring medical treatment, potentially preventing workers from performing physical tasks.

WMSDs develop over time due to non-ergonomic working conditions and can be categorized into two main groups:

- Back disorders related to handling loads
- Upper limb disorders, also known as repetitive strain injuries (RSIs)

MSDs can be caused by, for example:

- Twisting and bending movements while handling loads
- Repetitive or forceful movements
- Awkward and static postures
- Exposure to vibrating tools
- Prolonged sitting or standing in the same position

An increase in MSDs has been reported worldwide. This rise is partly attributed to improved recognition and reporting, but also to the tendency for work to be more segmented and specialized. This trend is driven by an increasingly competitive global economy, where fast-paced repetitive work environments are more widespread.

1.1.2 Industrial motivation for ergonomic improvement

Musculoskeletal disorders affect approximately one in three people worldwide. Within the European Union, MSDs represent the most costly health-related problems, affecting 45 million workers. Many individuals are unable to continue working beyond the age of 60 due to these conditions, resulting in billions of euros in lost revenue and a substantial burden on the public healthcare system.

A declining physical condition can also lead to mental health issues such as depression or frustration, resulting in reduced job satisfaction and, indirectly, lower work performance, even when workers are still physically able to continue working.

The increase in education in the workforce, combined with higher expectations regarding job satisfaction, has reduced workers' tolerance for WMSDs. An ergonomic work environment can prevent the development of WMSDs and directly reduce both the cost to companies due to sick leave and the societal impact.

Reduced work ability can be categorized into absenteeism and presenteeism. Absenteeism refers to the absence from work due to injuries sustained on the job. Presenteeism describes the situation in which workers continue to attend their expected work duties despite existing injuries, often due to concerns about job security. In such cases, work is performed in a self–protective manner to reduce physical strain.

In Belgium, 2.5 million people were affected by MSDs in 2018 (approximately 62% of the Belgian working population in 2015), resulting in 3 billion euros in direct medical costs and 2 billion in work absenteeism, corresponding to 0.3% of Belgium's gross domestic product [1][2][3][4][5].

1.1.3 Physical human–robot interaction

There is a clear need to address reduced work efficiency resulting from physical injuries since this improves the quality of life of an increasingly older workforce, while also reducing the impact on social welfare systems and increasing overall economic productivity. Industry has already adopted robotic solutions to achieve accuracy, speed, and repeatability that human workers cannot provide. Humans, however, provide flexibility and have the capability to react to unforeseen circumstances. By increasing the degree of human participation, the strength and repeatability of robots are combined with the adaptability and decision–making skills of humans. Humans and robots each operate in their respective workspaces to perform their tasks, and in some cases share the same workspace, as illustrated in Figure 1.1.

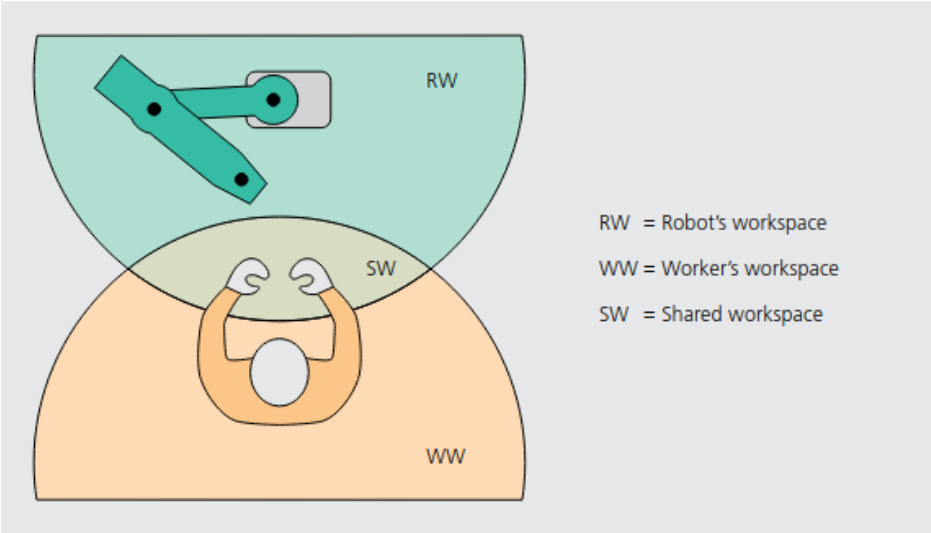

Figure 1.1: Human, robot, and shared workspace from Fraunhofer [6].

Human–robot cooperation generally refers to the use of robots without safety fencing, also called cage–free robots. This is in contrast to traditional robots which are stiff, heavy, and lack safety features that allow them to operate in the same space as humans, and are consequently only allowed to operate in closed –off workspaces.
Humans and robots can work together in different types of operation, as illustrated in Figure 1.2.

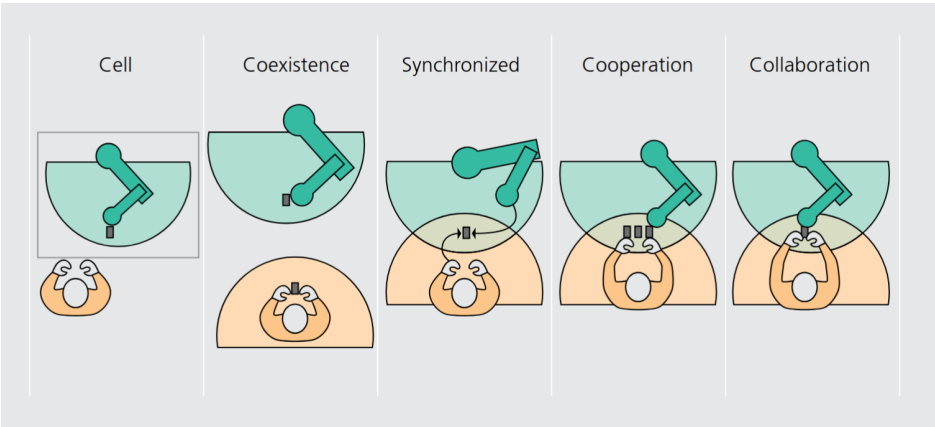

Figure 1.2: Levels of cooperation between human worker and robot from Fraunhofer [6].

From left to right an increasing level of interaction is shown.

- Cell: Robot operates in traditional cage, fenced–off from human contact
- Coexistence: Human and cage–free robot work at the same time, but in different workspaces
- Synchronized: Humans and cage–free robots work in the same workspace, but one at a time
- Cooperation: Human and cage–free robot have tasks in the same workspace at the same time, but do not work together on the same task

- Collaboration: Human and cage–free robot work in the same workspace on the same task

The robotic testbench developed in this thesis operates in collaboration mode during assisted load handling, when the human operator physically guides the robotic arm, and in coexistence mode during the floating stand–by state, when no physical interaction is taking place. This distinction is further clarified in chapter (idk exactly for now).
Collaborative robots, defined as robots capable of performing collaborative tasks as described above, are widely used in industry. However, they are still primarily applied in traditional robotic operations such as pick–and–place tasks. In many cases, humans and robots remain either physically separated or operate in a coexistence mode, where no physical barrier is present but no direct interaction takes place. Such implementations are only economically viable for large batch sizes, since a whole production line needs to be tuned towards that specific robotic solution.
Cage–free robotic arms that share a workspace with a human worker are introduced to increase production flexibility. Interactive robotic solutions for small batches and varying product sizes have become a clear need since 70% of collaboration between human workers and robots occurs in assembly applications. An industry survey reported that 86% of respondents were willing to invest in robotic solutions, indicating strong industrial interest. The primary motivations for introducing robotic arms in assembly environments are:

- Improving operational efficiency
- Improving ergonomics

Cage–free robotic solutions generally involve higher initial costs compared to traditional fenced systems due to safety certification requirements and conformity procedures to minimize the risk of injuries. However, companies are often willing to accept longer payback periods for implementing and testing company–specific robotic solutions which increase ergonomics [6][7].
ISO/TS 15066 covers the safety requirements for collaborative operations between human workers and robots. This normative classifies four different collaborative operations modes [8][9]:

- Safety–rated monitored stop
- Hand guiding
- Power and force limiting
- Speed and separation monitoring

Safety–rated monitored stop

The robot operates at its rated speed when a human is not present in the collaborative workspace. When a human enters the workspace, which can be detected by various means like laser scanners or light curtains, the robot undergoes a complete stop. As long as the human is in the workspace, the robot will not operate. Only by leaving the workspace and resetting the safety stop, can operation continue.

Hand guiding

Hand guided robotic applications are defined as a direct physical interaction between humans and robots, where the human operator has full control over the robot's motion. In this approach, a direct input device is located at or near the end–effector to measure input forces and simultaneously define the position of the human in the shared workspace. The robot must be in a monitored stop before hand guiding can be activated via an enabling device. Safety requirements for this mode include force and/or torque sensing at the guiding point, emergency stop accessible from the position where the robot is being guided, which is typically at the end–effector, and a maximum speed limit during guided motion.

Power and force limiting

This mode is the most commonly associated with collaborative robots. The robot's software and hardware are designed in such a way that any contact with a human cannot result in injury. Low robot mass, rounded surfaces, and padding are combined with active force and torque sensing, which limits contact forces with the environment below defined thresholds depending on the body region of the human that can be in contact with the robot.

Speed and separation monitoring

Robots and humans can operate in the same workspace at the same time, but the position and speed of the human and robot relative to each other is constantly monitored to adjust the speed of the robot with the goal of constantly having a minimum dynamical separation distance at all times.

Hand guiding is useful for packaging and assembly applications since it combines the physical strength of robotic arms, which can compensate for the weight of components or help with move relatively light but large components, with human intelligence.
ISO/TS 15066 specifies that the hand–guiding devices should be located at or near the robot end–effector, structurally restricting compliant interaction to a single designated contact point. Collisions or attempted handling of the robotic arm elsewhere will result in a safety stop [10].
Existing robotic solutions that allow a human worker to hand guide a robotic arm while simultaneously identifying and compensating for payloads of unknown variable weights, without requiring prior software configuration, are to the best of the author's knowledge, not available in assembly and packaging industries. When hand guiding is discussed in literature, it is typically limited to sensorized end–effectors on conventional stiff robots. Collaborative robots adopted in industry generally lack build–in compliance. As previously stated, collaborative robots are used for applications with little to no human contact, where precision and repeatability is more important than inherent compliance [11][12].
This absence in current industrial applications is further supported by surveys of end users, system integrators and researchers, which indicate that collaborative robots are still primarily used as conventional stiff robots with limited or no direct human interaction, while offering the added value of less space requirements for operation [13].

Current collaborative robots with the highest payloads, such as the UR30 from Universal Robots and the CR-35iB from FANUC, can handle loads up to 30kg and 50kg respectively. Still, hand guiding is mainly used for point– and path–based teaching and not for weight compensation tasks in hand guiding applications. The user can intuitively move the robot to a new point in space and record that position. This can be done for a single point or for a full path; in this case the robot records the continuous movement while a button is pressed. It is, however, possible to correctly configure collaborative robots for weight compensation, but software is required for correct calibration of the load attached to the end–effector. Although rare, some collaborative robots can estimate payload weight and center of mass by performing predefined controlled motions, but multiple interactions with the software controller are needed, making this approach time–consuming. This method is not generally available, and is often part of an optional software package that is not compatible with every collaborative robot. This confirms that automatic real–time payload identification without software interaction is not a standard feature of current collaborative platforms deployed in assembly and packaging contexts industries, representing a key limitation addressed by this thesis [14][15][16][17].

1.1.4 Weight assisting devices

Passive weight compensating manipulators are well established in industrial environments to help human workers perform their tasks in a more ergonomically sustainable way. These devices, if well calibrated, allow payloads to be moved around with minimal effort, offering passive, purely mechanically sourced weight compensation, effectively combining software–free and intuitive use. These manipulators however share a fundamental limitation; they only operate efficiently only within a pre-tuned weight range. The further a payload differs from the equilibrium point, the more force the human operator will have to deliver to move the load fixed to the manipulator. As a direct consequence, varying payloads that are too different in weight cannot be manipulated with these devices without manually re–tuning the counter–force system, for example counterweights and gas springs [18]. The proposed system replicates the intuitive, software–free interaction properties of a passive balancer, while adding real–time load identification and the possibility to perform autonomous tasks, integrating this solution into the production line. Some commercially available examples support this claim.

The company 3ARM produces commercially available weight compensating manipulators, an example of which shown in Figure 1.3. Pneumatic gas springs are used to compensate weights within a minimum and maximum payload limit. The gas springs can be manually tuned to compensate for different weight classes [19]. Figure 1.4 presents an example of a gas spring characteristic. By adjusting the filling pressure, the spring can optimized for different payloads scenarios [20].

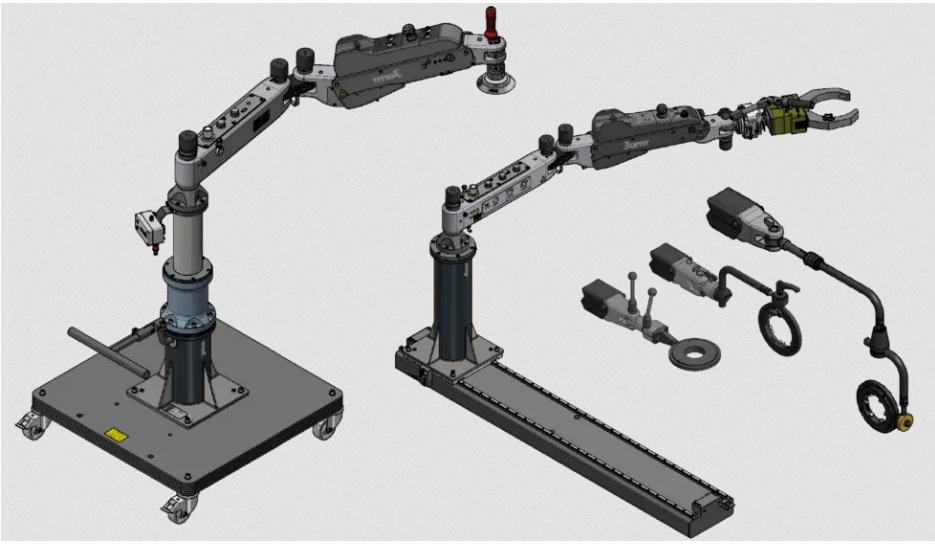

Figure 1.3: 3ARM manipulator [19].

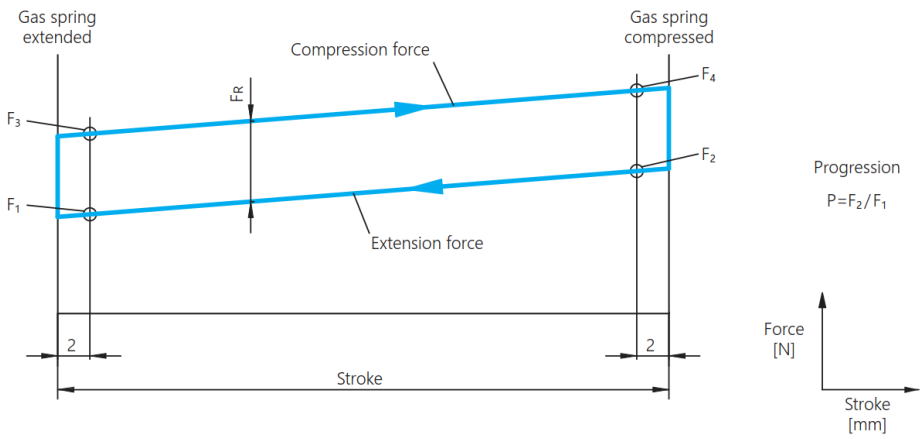

Figure 1.4: Gas spring characteristic by Suspa [20].

Transprotec offers spring balancers, showh in Figure 1.6, that are tuned to a specific weight, load changes are not permitted to guarantee safety and correct operation [21].

Figure 1.5: Spring balancer from Transprotec [21].

1.1.5 Intelligent Assist Devices

The robotic testbench proposed in this thesis belongs to the category of Intelligent Assist Devices (IADs). These are described as computer–controlled, servo–driven tools that enable human workers to manipulate payloads in a safe and ergonomic way. This differentiates IADs from the devices described in Section 1.1.4, which were completely passive. IADs were introduced to increase productivity and mitigate non–ergonomic work environments, a major issue in societies with an aging workforce. Currently, the term IADs is not commonly used anymore, as it has been subsumed under the modern term collaborative robots [22]. When IADs are mentioned today, they primarily appear as overhead gantry–mounted installations. While IADs can be effective in dedicated production cells with sufficient overhead clearance, they are unsuitable for industrial environments where ceiling height is limited, or overhead obstructions are present.

IADs cannot only interact safely and compliantly with human operators, but can also actively follow a predefined path. An example of this semi–autonomous strategy can be found in the automotive industry, where the controller guides the payload along a moving assembly line so that the human operator can focus on correctly installing the part into the assembly, rather than continuously repositioning the payload, as shown in Figure 1.6[23][24].

Figure 1.6: IAD integrated in an assembly line [24].

1.2 Problem Statement

Non–ergonomic load handling is the primary cause of musculoskeletal disorders in packaging and assembly industries. Passive weight compensating devices are well established in industry with the goal of reducing this physical strain on the human body; these devices however cannot adapt to varying payloads and provide no sensing, intelligence, or active integration with production lines.
Intelligent assist devices offer semi–autonomous advantages to support payload manipulation. Gantry–style implementation cannot be used in buildings with limited ceiling heights or overhead obstructions. IADs are semi–autonomous solutions and can thus offer a higher degree of integration than passive devices.
Collaborative robots require frequent software interaction to calibrate the system for different payloads or work modes. Although rare, it is possible for specific collaborative robots to go through predefined motions to estimate the weight of a payload rigidly attached to the end–effector, which also requires software programming and confirmation. Payloads also need to be fixed to the end–effector, which is not always possible for loads that vary in shape and size.
This thesis presents a three degree of freedom robotic testbench using novel, inherently compliant transmissions. This allows for the combination of an intrinsically safe design with real–time weight measurements and intuitive, software–free hand guidance at any point along the arm structure. This approach aims to combines the strengths of collaborative robots with passive or intelligent assist devices.
The testbench provides a weight–optimized platform that allows for further validation of the novel transmission, offering data acquisition and a configurable controller written in C++ specifically for the approach of this thesis.

1.3 Thesis Outline

In Chapter 2, the theoretical background on concepts relevant to this thesis is introduced, followed by an overview of the current state of the art in collaborative load–handling systems.

In Chapter 3, the design of the robotic testbench is presented; this includes the mechanical design, hardware implementation, proposed controller, and data communication architecture.

In Chapter 4, the developed testbench and proposed controller are experimentally validated; this includes mechanical design validation and controller performance evaluation in interactive load–handling experiments.

Chapter 5 summarizes the main conclusions of this thesis, discusses limitations of the proposed system, and presents possible improvements for future development.

Chapter 2

Literature Review

2.1 Theoretical Background

2.1.1 Impedance control

Impedance control is widely used in applications that involve physical human–robot interaction. The interaction forces between the robotic structure and its environment can be regulated by adjusting the apparent mechanical impedance of the robot. Mechanical impedance is defined as the ratio of the output force to the input velocity. It represents how much the robotic structure resists motion when an external force is applied.

$$Z(s) = \frac{F(s)}{v(s)} \tag{2.1}$$

Its inverse, mechanical admittance is defined as the ratio of input velocity to the output force.

$$Y(s) = \frac{v(s)}{F(s)} = \frac{1}{Z(s)} \tag{2.2}$$

When two mechanical bodies interact, the interaction can be characterized by their relative mechanical properties. A body that resists motion by responding to displacement with forces is modelled as a mechanical impedance. A body that responds to forces by generating motion or deformation is modelled as a mechanical admittance. These roles are not inherent properties of the objects themselves, but a consequence of the mechanical properties of the interacting bodies relative to each other, and the choice of input and output signals.

In conventional robotic applications, the robotic structure is stiff, position–controlled, and has a high inertia compared to its environment, like a human worker or payloads. It follows a defined trajectory and resists external forces; in this scenario, the robotic structure behaves as an impedance. The environment, whether it is a workpiece, or a human worker, yields to externally applied forces and is typically modelled as an admittance. As an example, pushing a trolley or compressing a pillow by applying an external force can clarify this concept; in both cases the environment moves or deforms in response to an applied force.

The impedance or admittance assignment is context–dependent. A rigid concrete wall presents an impedance that is much higher than that of a lightweight robotic structure; from the perspective of the robotic structure, it must yield to its much stiffer environment.

A human operator physically guiding a robotic arm applies forces and expects the robotic structure to move; the human worker behaves like an impedance, while the robotic structure behaves like an admittance. Impedance control modifies the mechanical impedance of a robotic structure to control how it behaves relative to its environment.
Figure 2.1 illustrates a stiff robotic structure interacting with its environment. The schematic at the end–effector demonstrates how impedance control can modify the apparent impedance.

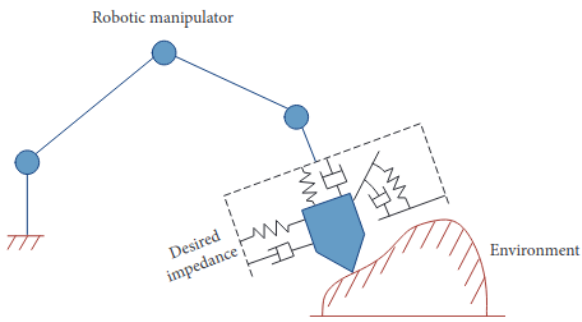

Figure 2.1: Impedance control for robotic structure in contact with an environment [25].

Impedance control mirrors the ability of the human body to change its mechanical impedance by tensioning its muscles. To guarantee safe interaction between a robotic structure and its environment, one of the two bodies should be compliant relative to the other. Most collaborative robots lack the ability to dynamically change their mechanical impedance; therefore, safe interaction between between a robot and an unknown environment cannot be guaranteed by mechanical design alone. Impedance control allows for a user–defined, dynamic relationship between the trajectory of the end–effector and the contact forces from the environment, allowing stiff robotic structures to compliantly react to their environment.

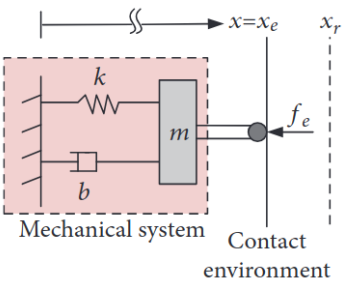

Figure 2.2: Mass–spring–damper system in contact with external force f_e [25].

The mathematical derivation for impedance control starts by considering a second–order system as shown in Figure 2.2.

$$m\,\ddot{x} + b\,\dot{x} + k\,x = u + f_e \tag{2.3}$$

In this equation, m represents the system mass, b the damping coefficient, k the stiffness coefficient, u the control input (as seen from the plant's perspective) generated by the controller which is applied to the plant dynamics, and f_e is the external force applied to the system.

The position $x = x_e$ represents the position of the contact surface and can be interpreted as the position error relative to the equilibrium position x_r.
As stated before, the idea of impedance control is to change the apparent impedance of a mechanical system. By defining a target impedance, the mechanical impedance of the system can be modified. This target impedance is expressed as a virtual mass–spring–damper system:

$$m_d\,(\ddot{x} - \ddot{x}_r) + b_d\,(\dot{x} - \dot{x}_r) + k_d\,(x - x_r) = f_e, \tag{2.4}$$

In this equation, m_d, b_d, and k_d respectively represent the desired inertia, damping, and stiffness of the system. A lower k_d makes the robotic structure more compliant, while a higher k_d makes it stiffer. The damping b_d opposes velocity and influences how fast the system will settle. The inertia m_d determines the system's resistance to acceleration.
A general classification of impedance control approaches is shown in Figure 2.3. The conventional approaches are discussed in this thesis, as these provide a sufficient conceptual understanding of the topic [25]–[28].

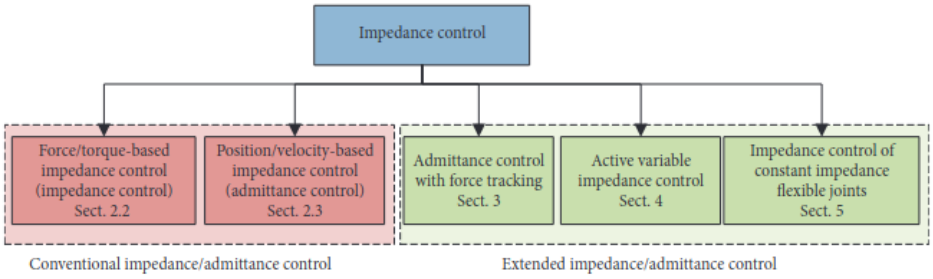

Figure 2.3: General classification of the impedance control approaches [25].

Force–based impedance control

In force–based impedance control, the control input u is a force or torque generated by the controller and is applied to the plant. Consequently, the motor drivers should be able to directly control their output torque; motor drivers only capable of position or velocity control cannot directly implement an impedance controller.
The control law is derived by requiring that the system satisfies the target impedance of Equation (2.4). In the first step, the plant dynamics (2.3) are solved for the system's acceleration term $\ddot{x}$:

$$\ddot{x} = \frac{1}{m}(u + f_e - b\,\dot{x} - k\,x). \tag{2.5}$$

The target impedance (2.4) is solved for the desired acceleration $\ddot{x}$:

$$\ddot{x} = \ddot{x}_r + m_d^{-1}(f_e - b_d\,(\dot{x} - \dot{x}_r) - k_d\,(x - x_r)). \tag{2.6}$$

For the system to satisfy the desired impedance behaviour, the actual acceleration (2.5) and the desired acceleration (2.6) must equal each other.

$$\frac{1}{m}(u + f_e - b\,\dot{x} - k\,x) = \ddot{x}_r + m_d^{-1}(f_e - b_d\,(\dot{x} - \dot{x}_r) - k_d\,(x - x_r)). \tag{2.7}$$

Multiplying both sides by m and solving for u:

$$u = b\,\dot{x} + k\,x - f_e + m\,\ddot{x}_r + m\,m_d^{-1}(f_e - b_d\,\dot{x} + b_d\,\dot{x}_r - k_d\,x + k_d\,x_r). \tag{2.8}$$

Collecting terms in $\dot{x}$, x, and f_e gives the force–based impedance control law as presented in Equation (2.9).

$$u = (b - m\,m_d^{-1}\,b_d)\,\dot{x} + (k - m\,m_d^{-1}\,k_d)\,x - (1 - m\,m_d^{-1})\,f_e + m\,m_d^{-1}\,(b_d\,\dot{x}_r + k_d\,x_r + m_d\,\ddot{x}_r). \tag{2.9}$$

When desired inertia m_d equals the actual inertia m, the inertia ratio $m\,m_d^{-1}$ becomes unity. This results in the external force term f_e being canceled, and Equation (2.9) simplifies to:

$$u = (b - b_d)\,\dot{x} + (k - k_d)\,x + b_d\,\dot{x}_r + k_d\,x_r + m\,\ddot{x}_r. \tag{2.10}$$

The elimination of f_e from the control law holds only when $m = m_d$ exactly. In practice, the effective inertia of a robotic structure cannot be perfectly matched by a fixed nominal value, resulting in behaviour that does not fully conform to the simplified control law. Since the term representing the external force has been canceled, no force sensor is required. This formulation represents a velocity and position feedback with feed–forward desired trajectory values.

An important limitation of this approach is that contact force is only controlled indirectly by specifying a reference position. For example, for a rigid robotic structure in free space, the stiffness k is considered $0\,\mathrm{N\,m^{-1}}$ and the desired spring constant k_d is set to $500\,\mathrm{N\,m^{-1}}$. If a contact force of 10 N is desired, the deviation from the reference position must be exactly 20 mm. This cannot be guaranteed unless the physical properties and position of the environment are exactly known, which is not always feasible. As a consequence, impedance control can either guarantee accurate position tracking or force tracking. A stiff value for k_d will produce high forces for small position errors resulting in accurate position tracking. A soft spring will result in accurate force control, but with large position errors as a trade–off. A diagram of impedance control is shown in Figure 2.4. Note that only a reference position is used; whether reference velocity and acceleration are also considered depends on the application and the available measurements [25]–[28].

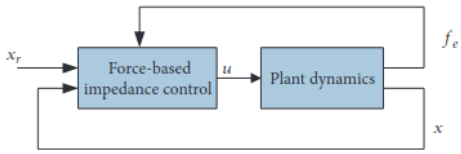

Figure 2.4: Impedance control diagram, with reference position x_r, external force f_r which in theory can be canceled, and a measured position x [25].

Position–based impedance control

Position–based impedance control is commonly referred to as admittance control, and is, as previously stated, the inverse of impedance control which was discussed above. Where impedance control reacts to motion deviation by generating forces, admittance control reacts to forces by deviating from the desired motion. Contrary to impedance control, a force measurement is required, meaning that sensors need to be fitted to the contact point with the environment.

A position–based impedance controller consists of two control loops as shown in Figure 2.5. The outer loop imposes the desired target impedance by generating a commanded reference position x_c, the inner position loop tracks that compliant reference position generated by the outer loop.

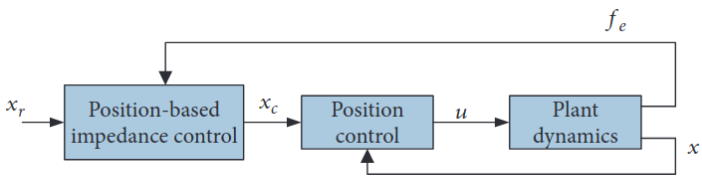

Figure 2.5: Admittance control diagram, with reference position x_r, compliant reference position x_c, measured position x, and measured external force f_e [25].

The outer loop requires both the reference position x_r and the measured external force f_e as input to compute the value for x_c as shown in Equation (2.11), where x_c is the only unknown variable.

$$m_d\left(\ddot{x}_c - \ddot{x}_r\right) + b_d\left(\dot{x}_c - \dot{x}_r\right) + k_d\left(x_c - x_r\right) = f_e. \tag{2.11}$$

When no external force is measured ($f_e = 0\,\mathrm{N}$), the commanded impedance reference position x_c must equal x_r. When the external force $f_e \neq 0$, x_c deviates from x_r according to the desired impedance parameters m_d, k_d, and b_d. For example, when an external force $f_e = 10\,\mathrm{N}$ is applied and $k_d = 500\,\mathrm{N\,m^{-1}}$, ignoring the velocity and acceleration terms at steady–state, a deviation of $x_c - x_r = 20\,\mathrm{mm}$ is calculated.

This commanded position is used as the input for the inner loop, which can be a conventional position controller such as a proportional–derivative (PD) controller as shown in Equation (2.12), which tracks the commanded trajectory x_c.

$$u = k_p\left(x_c - x\right) + k_v\left(\dot{x}_c - \dot{x}\right), \tag{2.12}$$

The inner loop has no knowledge of the desired impedance behaviour, its only purpose is

to track the commanded position as accurate as possible. Therefore it must be sufficiently stiff in order for the admittance controller to work as intended. If the position control loop is compliant, the actual robot position will not accurately follow x_c. Consequently, the impedance felt by the operator becomes a combination of the outer impedance loop and the inner position loop, possibly creating unsafe behaviour [25]–[28].

Position–based versus velocity–based inner loop

The inner loop can track either a commanded position x_c or a commanded velocity $\dot{x}_c$. This choice depends on the application, which can be illustrated by considering a step force applied to the robotic structure by the human operator. In collaborative load handling, there is no spring that pulls the robot arm back to a static reference position. The human operator moves the robotic structure freely and expects it to stay wherever it is released, instead of pulling back. Therefore, in this example, the desired spring constant $k_d = 0\,\mathrm{N\,m^{-1}}$ to ensure that the robotic structure to not pull back against the force applied by the operator. To illustrate the difference between a position inner loop and a velocity inner loop, consider Equation (2.11), simplified by setting the desired inertia term m_d and reference velocity $\dot{x}_r$ to zero.

$$b_d\,\dot{x}_c = f_e \tag{2.13}$$

The step force applied by the human operator f_e is constant after $t = 0^+$, resulting in Equation (2.14) and (2.15).

$$b_d\,\dot{x}_c = 1 \tag{2.14}$$

$$\dot{x}_c = 1/b_d \tag{2.15}$$

A constant force will result in a constant velocity $\dot{x_c}$. Consequently, the commanded reference position x_c grows linearly over time. This behaviour results from the absence of the virtual spring which would create an equilibrium position. With a position–based inner loop, the controller tries to track this continuously growing commanded position, meaning that any lag in the inner loop can cause accumulation of the tracking error. Therefore, this approach requires aggressive tuning, which is more prone to instability issues.

With a velocity–based inner loop, the controller tracks a commanded reference velocity $\dot{x_c}$ instead. Since $\dot{x_c}$ is constant for the theoretical step force applied by the operator, the controller tries to match a constant velocity instead of a continuously growing position. This approach allows for less aggressive tuning, and thus a more stable controller.

The mass–spring–damper system with the commanded trajectory x_c is shown in Figure 2.6. The admittance controller combines the measured force f_e with the reference position x_r. When the robotic structure should remain in continuous contact with the environment, such as in grinding applications, x_r can be set slightly beyond the contact surface to maintain a constant contact force [25]–[28].

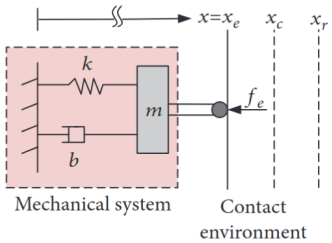

Figure 2.6: Mass–spring–damper system with commanded position x_c [25].

Comparison of Force-Based and Position-Based Impedance Control

Force–based impedance control commands torques directly and does not require an inner control loop. It can operate without a force sensor when the desired inertia is set equal to the actual inertia. To achieve stiff behaviour, high values for the impedance parameters must be chosen, amplifying the sensor noise which can result in instabilities. As was stated before, either the environment or the robotic structure should be compliant. If the environment is stiff, the robotic structure must feel soft, meaning lower values for the impedance parameters can be chosen resulting in less sensor noise amplification.
Admittance control requires both a force sensor and a stiff inner control loop. It is consequently better suited for stiff behaviour against compliant environments, such as a human operator. The performance depends on the quality of the inner loop and the force measurement. Most industrial manipulators are equipped motor drivers with integrated position controllers, and since admittance control uses an inner position loop, it can be implemented without redesigning the low-level control architecture [25]–[28].

2.1.2 Series elastic actuator

In conventional machine design, the interface between a motor and its load is made as stiff as possible. As discussed previously, robotic systems are still predominantly used for position-control tasks, where high structural stiffness is beneficial. A stiff structure improves position accuracy, stability, and bandwidth for position control.
When only the motor position is measured, a stiff structure minimises the position error between the motor shaft and the endpoint. For example, if a motor is connected to a rigid shaft as shown in Figure 2.7, the angular position of the endpoint can be accurately represented by the motor encoder reading. When this shaft is replaced by a compliant component such as a rubber coupling, as shown in Figure 2.8, the motor encoder no longer accurately reflects the angular position of the endpoint.

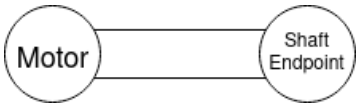

Figure 2.7: Motor and load connected with a rigid shaft.

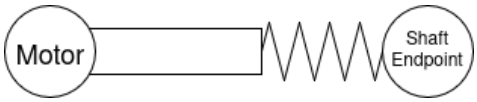

Figure 2.8: Motor and load connected with a compliant coupling.

When the position sensor is placed after the elastic coupling, the elasticity becomes part of the feedback loop. The motor and load inertia are coupled through the elastic element, and a stiffer coupling increases the resonance frequency of the system, allowing a higher control bandwidth:

$$\omega_n = \sqrt{k\left(\frac{1}{J_m} + \frac{1}{J_l}\right)} \tag{2.16}$$

Most electric motors have poor torque density: they can only achieve high power output at high rotational speeds. Since mechanical power is the product of torque and angular velocity.

$$P = \tau \cdot \omega \tag{2.17}$$

A lightweight, high-speed motor requires a transmission to deliver useful torque at the low speeds typical of robotic joints. Unfortunately, gears introduce significant drawbacks such as backlash, friction, torque ripple, and reflected inertia. For a gear ratio of N:1, the reflected inertia at the output is

$$J_{\text{reflected}} = J_{\text{motor}} \cdot N^2 \tag{2.18}$$

This N^2 amplification means that shock loads on the output shaft produce very high forces on the gear teeth, and it is often the load rating of the gears that limits the allowable output torque. The high reflected inertia also makes the actuator difficult to backdrive, which is problematic for force control and safe interaction with unknown environments. Because of low

torque density servo motors with a power rating much higher than the useful power output must be selected, resulting in expensive and oversized designs.

The series elastic concept

A solution to these problems was introduced by Pratt and Williamson, who proposed the Series Elastic Actuator (SEA). As the name describes, an elastic element is placed in series between the transmission output and the load, as illustrated in Figure 2.9.

Figure 2.9: General SEA schematic reconstructed from [29].

The spring obeys Hooke's law,

$$F = k \cdot \delta x \tag{2.19}$$

so by measuring the spring deflection δx and knowing the spring constant k, the force transmitted to the load can be calculated directly.
Masuring force through a stiff transmission is inherently difficult; friction, backlash, and torque ripple influence the relationship between motor current and output force. The series elastic element resolves this by converting the force measurement into a displacement measurement. The output force is determined by the spring deflection, which is a position that can be measured accurately regardless of the transmission's non–linearities. The motor receives a torque command, but the feedback signal that signals the controller whether the output force is correct is the spring deflection which is a position measurement. In this way, the force sensing problem is transformed into a position sensing problem. A force control problem consequently becomes a position control problem, which is the strength of motor equiped with a transmission.

Benefits of series elasticity

The spring acts as a mechanical low-pass filter on shock loads. When the output is subjected to a sudden impact, the spring deflects and spreads the impulse over a longer duration, reducing the peak forces on the gear teeth. Only the output–side mass contributes to the initial impact, the reflected inertia $N^2 J_{\text{motor}}$ is decoupled from the output by the spring. The lengthened impulse also gives the control system more time to accelerate the motor inertia and respond to the external force.
In a stiff actuator with force feedback, even a small displacement generates a large force reading. A high–gain feedback controller would respond aggressively, pulling back and causing the force to drop suddenly, resulting in chatter between the actuator and the load. To avoid this, the controller gains must be kept low, resulting in a slow system that is unable to respond to small forces.

With the introduction of a compliant element a moderate displacement generates a small force reading. This allows higher control gains without risking instability. At frequencies within the control bandwidth, the force–feedback loop can effectively cancel the spring force entirely, creating the behaviour of a zero-rate spring (an infinite compliant spring). From Equation (2.16), it is clear that when $k = 0\,\mathrm{N\,m^{-1}}$, the resonance frequency $\omega_n = 0\,\mathrm{rad\,s^{-1}}$. Since motor and load inertias cannot resonate with a zero–rate spring, the system is inherently stable when interacting with any passive object.

At high frequencies, the impedance of a traditional stiff actuator is dominated by its inertia, while a SEA presents the impedance of the spring, which is safer for interactions with unknown environments. However, since the high–frequency impedance of the SEA is governed by the spring, a light load can potentially resonate with it. Since the physical robotic structure has a mass, the resonant frequency of the system is lowered into te range where the control loop operates effectively thus being able to control apparent impedance of the system.

Because the force output of a SEA depends on the motor shaft position rather than the motor's internal torque, the transmission's non–linearities (such as friction, backlash, and torque ripple) do not directly affect the force measurement. These non–linearities affect the motor effort required to achieve a given shaft position, but since the force is measured via the spring deflection and incorporated in the feedback loop, the controller commands the correct motor motion regardless. This allows the use of less expensive gears and motors without sacrificing force–control quality.

Performance limits

The performance limits of a series elastic actuator can be understood by decomposing the required motor force. Consider the system shown in Figure **??**, where a motor with mass M_m is connected to the load through a spring with stiffness k_s.

The spring force on the load is given by Hooke's law:

$$F_l = k_s(x_m - x_l) \tag{2.20}$$

where x_m is the motor position and x_l is the load position. Newton's second law for the motor gives the net force to accelerate the motor mass:

$$M_m\ddot{x}_m = F_m - F_l \tag{2.21}$$

Rearranging:

$$F_m = F_l + M_m\ddot{x}_m \tag{2.22}$$

The motor position can be expressed in terms of the load position and the spring force by rearranging Equation (2.20):

$$x_m = x_l + \frac{F_l}{k_s} \tag{2.23}$$

Differentiating twice:

$$\ddot{x}_m = \ddot{x}_l + \frac{\ddot{F}_l}{k_s} \tag{2.24}$$

Substituting into Equation (2.22):

$$F_m = F_l + M_m \ddot{x}_l + \frac{M_m}{k_s} \ddot{F}_l \tag{2.25}$$

To analyze the system in the frequency domain, sinusoidal steady–state signals are assumed. Any sinusoidal signal can be represented using a complex exponential of the form:

$$x(t) = \hat{x} e^{j\omega t}, \quad F(t) = \hat{F} e^{j\omega t} \tag{2.26}$$

where $\hat{x}$ and $\hat{F}$ are complex amplitudes and ω is the angular frequency.
Using the complex exponential form allows derivatives to be computed easily. In particular, differentiating $e^{j\omega t}$ results in multiplication by $j\omega$, and the second derivative results in multiplication by $-\omega^2$.

$$\frac{d}{dt} e^{j\omega t} = j\omega e^{j\omega t}, \quad \frac{d^2}{dt^2} e^{j\omega t} = -\omega^2 e^{j\omega t} \tag{2.27}$$

Therefore, the second derivatives can be replaced as:

$$\ddot{x} \to -\omega^2 x, \quad \ddot{F} \to -\omega^2 F \tag{2.28}$$

Applying these substitutions to Equation (2.25) gives:

$$\boxed{F_m = F_l - \frac{M_m \omega^2}{k_s} F_l - M_m \omega^2 x_l} \tag{2.29}$$

The three terms have the following physical interpretation:

1. F_L: F_l is the force applied to the load coming from the spring.
2. $F_m s$: $\frac{M_m \omega^2}{k_s} F_l$ is force used to accelerate the motor mass, which in turn will cause spring deformation. This term is unique to the SEA.
3. $F_m i$: $M_m \omega^2 X_l$ is the force required to accelerate the motor mass to track a moving load. This term exist regardless of the presence of a series spring.

The force that is required by the motor is limited to $F_{\max}$, which in practise can be a limit on the motor current to prevent overheating for example.

$$|F_m| \leq F_{\max} \tag{2.30}$$

The motor force constraint $|F_m| \leq F_{\max}$ can be visualised as a vector diagram (Figure 2.10). The vector F_m is the sum of three components, and its magnitude must stay inside a circle of radius $F_{\max}$.

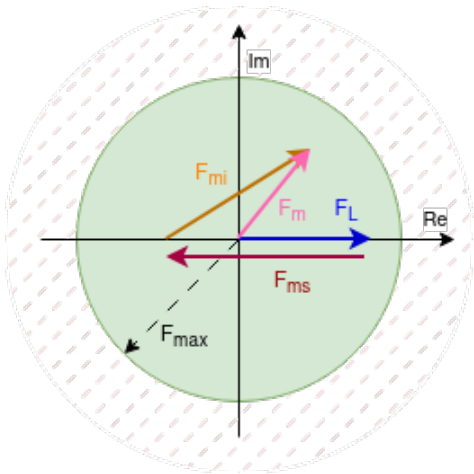

Figure 2.10: Vector diagram reconstructed from [29].

F_l is arbitrarely aligned with the real axis. The second component (the SEA term) points in the opposite direction along the real axis, moving the starting point of the third vector to $F_l(1 - M_m\omega^2/K_s)$. From this new starting point, the third component $(-M_m\omega^2 X_l)$ extends in the direction determined by the phase relationship between force and load motion. The final sum gives F_m.

The series elasticity is beneficial for all frequencies $\omega < \sqrt{2\frac{k_s}{M_m}}$ given by Equation (2.33), since the vector F_m gets smaller. The actuator can handle faster or larger load motions before saturating at $F_{\max}$.

$$\left|1 - \frac{M_m\omega^2}{k_s}\right| < 1 \tag{2.31}$$

Expanding the absolute value inequality:

$$-1 < 1 - \frac{M_m\omega^2}{k_s} < 1 \tag{2.32}$$

The right–hand side gives $\frac{M_m\omega^2}{k_s} \geq 0$, which is always satisfied since $M_m > 0$, $k_s > 0$, and $\omega^2 \geq 0$. The left–hand side gives:

$$\frac{M_m\omega^2}{k_s} < 2 \quad \Longrightarrow \quad \omega < \sqrt{\frac{2k_s}{M_m}} \tag{2.33}$$

The series elasticity improves or limits performance depends on the task. When the output force varies, the motor must move to change the spring deflection, and the motor's inertia and torque limits restrict how quickly this can happen and thus limit the bandwidth in which the actuator can guarantee a zero–rate spring. When a time–varying force $F(t)$ is commanded by generating motor motion defined by $x(t) = F(t)/k$, the load motion $x_l(t)$ can add constructive or destructive to the spring motion $x(t)$ depending on their relative amplitude and phase. When the load moves in a direction that naturally compresses the spring in the same direction as the desired force change, less motor motion is required and the effective bandwidth increases. When the load moves in the opposite direction, more motor motion is needed and bandwidth

decreases
The practical consequence for actuator design is that the spring stiffness should be chosen to match the task. For tasks requiring a constant force against a moving load, a softer spring causes small force errors and thus less motor effort. For tasks requiring varying forces on a stationary load, a stiffer spring is better because motor torque is transmitted more directly. No single spring stiffness is optimal for all tasks.

Control architectures

Pratt and Williamson proposed a control architecture that combines feedforward compensation with feedback correction. The architecture is shown in Figure 2.11, where the transfer functions are derived from Equation (2.29) by using $s = j\omega \rightarrow \omega^2 = -s^2$.

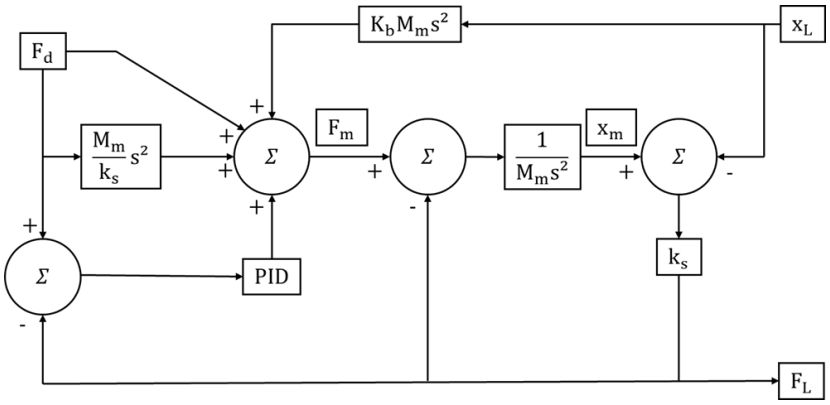

Figure 2.11: Torque control diagram with feedforward terms from [29].

The controller receives a desired output force F_d and the measured spring force $F_l = k_s(x_m - x_l)$. The controller's task is to make F_l track F_d.
The force error $F_d - F_l$ is processed by a PID feedback loop. Feedforward terms compensate for the components of the required motor force derived in Equation (2.29). The first two terms, the desired output force and the motor acceleration needed to change the spring deflection, are fully compensated in the feedforward path. The third term, motor acceleration to track load motion, is only partially compensated by choosing a $K_b < 1$ to prevent instability. The combined feedforward and feedback output is a desired motor current, which translates to a motor torque [29].

Yu et al. proposed a cascaded control architecture consisting of three nested loops [30]. The architecture is shown in Figure 2.12.

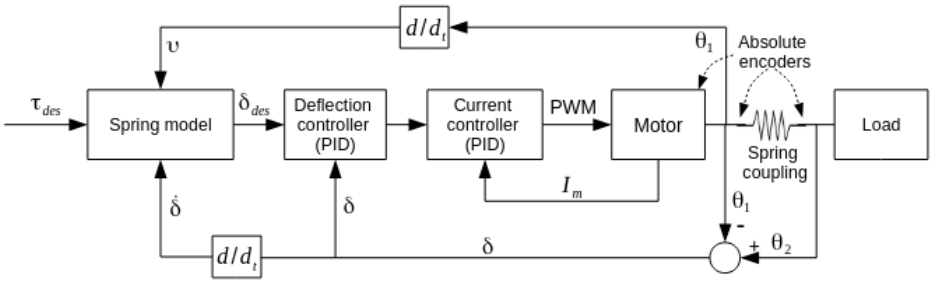

Figure 2.12: Torque control diagram from [30].

The outermost layer is the spring model, which converts the desired torque τ_{des} into a desired spring deflection δ_{des}. For a linear spring, this conversion is:

$$\delta_{\text{des}} = \frac{\tau_{\text{des}}}{K_s} \tag{2.34}$$

When the spring exhibits nonlinear behaviour such as hysteresis, a data–driven model can be used instead. Yu et al. created their own model to better model the spring behaviour.
The middle loop is a PID controller that drives the measured spring deflection δ towards the desired deflection δ_{des}. The spring deflection is computed as the difference between two absolute encoders placed on either side of the elastic element. The output of this deflection controller is a motor current command.
The innermost loop is a motor current controller (PID), which regulates the actual motor current to match the commanded value.
Compared to the Pratt and Williamson architecture, the cascaded approach does not use explicit feedforward compensation for motor inertia or load motion. Instead, these effects are handled implicitly by the deflection PID. If friction or inertia causes the actual deflection to deviate from the desired value, the deflection controller increases the motor current until the error is corrected. The absence of feedforward terms makes the controller simpler to implement, at the cost of relying entirely on the feedback gains to reject disturbances.
[29]–[32]

2.2 State of the Art

Chapter 3

Methodology

3.1 System Overview and Requirements

3.2 Mechanical Design

3.3 Hardware Architecture

3.3.1 CAN Bus Topology

3.3.2 Sensor Architecture

3.4 Sensor and Actuator Modeling

3.4.1 SET Spring Model

3.4.2 Motor and Transmission Model

3.5 Control Architecture

3.5.1 State Machine

3.5.2 Gravity Compensation and Floating Mode

3.5.3 Weighing Mode

3.5.4 Carrying Mode

3.5.5 Put-Down Mode

3.6 Software Implementation

3.6.1 Real-Time Architecture

Example code listing:

```
while (running) {
    clock_nanosleep(...);
    // your code here
}
```

Listing 3.1: Example listing.

3.6.2 Configuration

Chapter 4

Validation and Results

4.1 Experimental Setup

4.1.1 Hardware Configuration

4.1.2 Parameter Identification

4.2 Spring and Sensor Characterization

4.3 Controller Performance

4.4 Assisted Load Handling Evaluation

4.5 Discussion

Example table:

Table 4.1: Example results table.

Metric	Target	Achieved
Value 1	X	Y
Value 2	X	Y

Chapter 5

Conclusion and Future Work

5.1 Conclusion

5.2 Future Work

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2D Manufacturing Drawings

.1 CNC Part 1

.2 CNC Part 2

.3 CNC Part 3

Additional Experimental Results

Source Code

https://github.com/username/repository